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February 21, 2026

CSL7360 - Computer Vision

Instructions:
1. Read the questions carefully.
2. All questions are mandatory.
3. If you find anything unclear/incorrect in any question, make a reasonable assumption and proceed.

Time: 2 hour | *Minor* | Maximum Marks: 30

1. Which of the followings is/are true for affine transformations. [2]

   (I) A line passing through origin necessarily passes through the origin after transformation.

   (II) The transformations are closed under composition.

   (III) Two lines with identical slope will have matching slopes after transformation.

   a) I and II b) I and III
   c) II and III d) All are true

2. Arrange the following pixels, with the given R, G, B values, in ascending order of their grayscale intensity. Assume that standard (luminance based) RGB to grayscale conversion is used. [2]

   a) 100, 50, 200
   b) 70, 100, 100
   c) 100, 70, 150
   d) 75, 75, 150

   a < d < c < b

3. What will be the effect of convolving a typical grayscale image with the following filters? [2]

   I) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

   II) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

4. Which among the followings will be the best filter for denoising a grayscale image corrupted with a) Gaussian noise, b) salt and pepper noise (scattered black and white dots)? Justify your answer. [2]

   bilateral median

   I) Gaussian filter II) Bilateral filter III) Median filter

5. How does SIFT achieve rotation and scale invariance? [4]

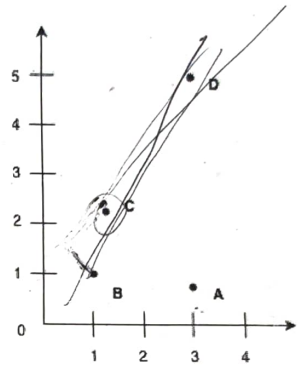
6. Recall the $2N \times 9$ matrix $\boldsymbol{A}$ created using matching keypoint pairs while estimating homography matrix parameters. Show that these parameters are same as the components of the eigenvector corresponding to the smallest eigenvalue of $\boldsymbol{A}^T\boldsymbol{A}$. [4]

7. Consider a homography matrix shown below with two missing elements [4]

$$\begin{vmatrix} 10 \\ 20 \\ 1 \end{vmatrix} = \begin{bmatrix} 1 & - & 1 \\ 0.1 & 1 & 0.5 \\ 0.1 & - & 1 \end{bmatrix} \begin{vmatrix} 2.5 \\ 3.58 \\ 1 \end{vmatrix}$$

If this transforms a point (10, 20) to (2.5, 3.58) find the values of missing elements.

8. Consider the figure shown on the right. It contains four points on a 2D plane **A**, **B**, **C**, and **D**. If we apply RANSAC algorithm to find the best-fit 2D line, which two points will lie on that line and why? [4]



9. Recall the three R's (recognition, reconstruction and re-organization) of computer vision discussed in the class and answer the followings. [6]

(I) Explain the unification three R's.

(II) Consider the following scenarios and explain which **R** is helping in which **R** (for example Re-organization helping in Recognition)

(a) RGB-D sensors provide a depth image in addition to the RGB image. It is observed that local gradient extracted from depth images help in improving the contour (object boundaries) detection task.

(b) An image is divided into overlapping patches for improving object detection.